Adarsha Pant

Email | Github | Portfolio | LinkedIn | Kathmandu, Nepal

Summary

Computer Engineering undergraduate and Software Engineer passionate about building scalable systems and integrating AI into web applications. Deeply interested in **Computer Vision, Machine Learning Algorithms, and System Design**. Experienced in bridging applied machine learning with production infrastructure, from training custom neural networks to architecting high-concurrency backend pipelines. Constantly experimenting with advanced ML algorithms to deploy intelligent, end-to-end web solutions.

Education

Kathmandu University **2022 – Present**
Bachelor of Engineering in Computer Engineering

V.S. Niketan College **2020 – 2022**
+2 in Science | GPA: 3.63/4.0 (A+)

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, SQL, PHP

Backend & Systems: FastAPI, Node.js, PostgreSQL, MySQL, SQLAlchemy, Alembic, Docker, JWT, WebSockets (Socket.io), Linux, REST API design

AI & Machine Learning: PyTorch, TensorFlow, OpenCV, YOLOv8, MediaPipe, Scikit-Learn

Frontend & Tooling: React, Next.js, Redux, NumPy, SciPy, Git

Projects

Signly | GitHub *Python, TensorFlow, PyTorch, MediaPipe, NLTK*

- Engineered a sign-language translation platform with live webcam gestures to English text and English text to sign-language video, unifying computer vision and NLP in one pipeline.
- Designed a custom spatio-temporal neural network (Conv1D feature extractor + Bidirectional LSTM) processing 30-frame skeletal-landmark windows, reaching 88.5% classification accuracy.
- Built preprocessing pipelines applying IQR and median scaling to normalize multi-dimensional MediaPipe landmarks for robust, user-agnostic gesture recognition.
- Developed an English-to-ASL grammar resolver using NLTK WordNet with custom parsers for contractions, spatial-index mapping, and token reordering into correct ASL syntax.
- Curated the WLASL training corpus via multithreaded scrapers (yt-dlp) with automated frame extraction and class balancing, deployed the pipeline behind a FastAPI service.

Road Damage Detection (SHP) | GitHub *Python, YOLOv8, OpenCV, PyTorch, FastAPI*

- Trained a custom YOLOv8 model to detect and localize 4 classes of road-surface defects on a 47,000+ image dataset spanning 6 countries.
- Built a data pipeline for dataset auditing, stratified train/val/test splitting, and augmentation (mosaic tiling, brightness/contrast, rain simulation) to improve generalization.
- Designed a Severity Index algorithm translating bounding-box geometry and confidence scores into real-world maintenance grades (Good/Fair/Poor/Critical).
- Bridged perception and planning by implementing A* search to auto-schedule repairs by cost, severity, and traffic weight.
- Deployed FastAPI inference endpoints that decode base64 images, run inference, and return OpenCV-annotated results in real time.

Code Room | GitHub *Node.js, Docker, Socket.io, MySQL, React*

- Engineered a sandboxed compiler API using Docker containers and Node.js child processes to securely execute untrusted JavaScript, Python, and C++ submissions.
- Built the real-time synchronization engine (WebSockets/Socket.io) powering multi-user collaborative editing and code execution with integrated chat.

- Implemented a custom auth flow (JWT, bcrypt, OTP email verification via NodeMailer) backing a session-persistent multi-user architecture.
- Integrated the Monaco Editor into a React/Vite frontend, with Redux Toolkit managing complex collaborative state.

IntelliMark AI Backend | GitHub *FastAPI, PostgreSQL, SQLAlchemy, Docker*

- Built an asynchronous FastAPI backend on PostgreSQL, modeling relational schemas for users, events, sponsors, and role-based access control.
- Designed a thread-safe task queue using OS-level file locking (`os.O_EXCL`) to safely schedule memory-intensive Stable Diffusion workloads on limited hardware without OOM crashes.
- Built an automated reporting engine (fpdf2) compiling live database metrics and AI-generated commentary into downloadable executive PDF reports.
- Implemented geospatial tracking with Leaflet, OpenStreetMap reverse-geocoding, and OSRM for distance calculation and turn-by-turn navigation.
- Managed schema evolution with Alembic, resolving migration branch conflicts and optimizing indexes for faster text search.

Space Cleaners | Live Site *PHP, MySQL, Gemini AI, React*

- Developed a custom PHP backend with JWT authentication, rate limiting, and an asynchronous cron-based email queue powering a live commercial platform.
- Shipped a full-stack platform combining a high-converting public website with an internal CRM for customers, orders, and automated invoicing.
- Integrated a context-aware customer-support chatbot powered by the Google Gemini 2.5 API.
- Designed a React/Tailwind frontend with a multi-step quote builder, dark mode, and Recharts-based admin analytics dashboard.

Additional Projects

Spashta Sound | GitHub *Python, FastAPI, NumPy, SciPy*

- Engineered a digital signal processing (DSP) pipeline entirely from scratch using NumPy/SciPy to perform Short-Time Fourier Transform (STFT), Spectral Subtraction, Wiener Filtering, and Overlap-Add reconstruction.
- Built a FastAPI backend for millisecond-latency audio processing and matplotlib spectrogram generation, serving a React dashboard with synchronized dual-track playback and interactive parameter tuning.

LL(1) Compiler Visualizer | GitHub *Python, FastAPI, React*

- Built a full-stack educational tool visualizing top-down predictive parsing for context-free grammars, with algorithms for left-factoring, left-recursion elimination, and FIRST/FOLLOW computation.

AlgoVision | GitHub *Python, PyQt6, Algorithms*

- Built an interactive desktop app visualizing execution, time complexity, and memory state for 6 sorting algorithms; used Python generators to drive a multithreading-free play/pause state machine.

Extracurricular Activities & Languages

Involvement: Technical Team Member, IT-MEET 2025 Annual technical event organized by the Dept. of Computer Science.

Languages: English, Hindi, Nepali